

December 2025

VT3101-1 – CMOS VLSI DESIGN
(For VDT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **NMOS pass-transistor pass** _____
A) Weak 0
C) Strong 0
B) Weak 1
D) Strong 1
2. **Transmission gate is a** _____
A) CMOS inverter
C) Unidirectional CMOS switch
B) CMOS buffer
D) Bidirectional CMOS switch
3. **The minimum number of MOSFETs required for the CMOS circuit of $y = \text{NOT}(ab + \bar{a}bc + a\bar{b}\bar{c})$ is** _____
A) 2
C) 6
B) 4
D) 8
4. **In the layout diagram of an NMOSFET, width of polysilicon is 180 nm and width of n+diffusion is 0.9 μm . Then the aspect ratio of the NMOSFET is** _____
A) 0.9/180
C) 90/180
B) 0.9/0.18
D) 90/180
5. **Which of the following is not true for a symmetrical inverter?**
A) $(W/L)_n = (W/L)_p$
C) $t_r = t_f$
B) $\beta_n = \beta_p$
D) $V_M = \frac{V_{DD}}{2}$
6. **Y – chart is useful in understanding**
A) Area and delay optimization
C) VLSI design flow and methodology
B) Power and delay optimization
D) Process technology and yield
7. **The spatial variation in arrival time of a clock transition on an integrated circuit is commonly referred to as** _____
A) Electrical effort
C) Path delay
B) Logical effort
D) Clock skew
8. **As β_n/β_p increases the CMOS inverter VTC shifts** _____
A) shifts to the left (toward Y axis).
C) remains unchanged.
B) shifts to the right (away from Y axis).
D) becomes symmetric about $V_{DD}/2$.
9. **Which CMOS logic uses differential signaling and complementary outputs?**
A) Static CMOS
C) CVSL
B) Pseudo-NMOS
D) NMOS Domino
10. **The number of transistors required for N input logic in Pseudo-NMOS logic is** _____
A) N
C) 2N
B) N+1
D) 2N+1

VT3101-1

SEE – December 2025



11. Hold time of a flip-flop is defined as:
 - A) Minimum time input must be stable after the clock edge
 - B) Minimum time input must be stable before the clock edge
 - C) Maximum time input can change after the clock edge
 - D) Time taken for output to respond
12. An unit inverter has input capacitance of C_{in} . If this inverter is scaled by a factor of 3 then the new value of input capacitance is _____.
 - A) C_{in}
 - B) $C_{in}/3$
 - C) $3C_{in}$
 - D) $6C_{in}$
13. The static CMOS logic structure suitable for complemented sum of products form of any logic function is _____.
 - A) AOI structure
 - B) OAI structure
 - C) Pass transistor logic structure
 - D) CVSL
14. Power consumed in CMOS digital circuits is mainly due to the _____.
 - A) PMOSFET
 - B) Bias current
 - C) Leakage current
 - D) Logic switching activity
15. The parity check detection is done using:
 - A) XOR gate
 - B) OR gate
 - C) AND gate
 - D) NOR gate
16. Which of the following CMOS logic is ratioed logic?
 - A) Pseudo NMOS
 - B) Dynamic CMOS
 - C) Static CMOS
 - D) CVSL
17. Test bench programs for HDL designs in Verilog are used to do _____.
 - A) Functional verification
 - B) Static timing analysis
 - C) Testing
 - D) Logic synthesis
18. Boundary scan test is used to test:
 - A) Pins
 - B) Multipliers
 - C) Boards
 - D) Wires
19. _____ is non-inverting because of the output inverter.
 - A) Dynamic CMOS logic
 - B) Static CMOS logic
 - C) Domino CMOS logic
 - D) Pseudo NMOS logic
20. The number of transistors required to construct N input function in footed dynamic CMOS logic is _____.
 - A) $2N$
 - B) $2+N$
 - C) $N+1$
 - D) $2N+1$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Sketch logic diagram and schematic in CMOS logic for (i) $Y = \overline{BC + A(D + E)}$ (ii) AOI 22 gate.	8	L*3	1	1
	b) Explain the significance of design rules in the layout design of CMOS digital circuits. With neat diagrams explain the λ design rules for diffusion, polysilicon and metal.	8	L2	2	1
2.	a) With minimum number of transmission gates realize (i) 4:1 MUX (ii) $y = AB + \overline{A}\overline{C} + A\overline{B}C$.	6	L3	1	1
	b) With neat diagram for cross sectional view and associated mask, explain the fabrication of CMOS inverter in p-well process.	10	L2	2	1
3.	a) With neat diagrams, explain regularity and modularity in VLSI design.	6	L3	1	1
	b) Draw the stick diagram and layout for $Y = \overline{AB + CD + E}$. Construct the graph of the network, identify the Euler path to find an input ordering which gives optimized layout.	10	L3	2	1

Unit – II

4. a) Sketch the DC characteristics of CMOS inverter and derive the expression for midpoint voltage of it. Deduce the condition for midpoint voltage to be $\frac{V_{DD}}{2}$.
- b) Write gate level diagram for negative edge triggered CMOS D-flip flop and explain the working.
5. a) Identify the values of x, y and z in the multistage logic network shown in Figure 5a.

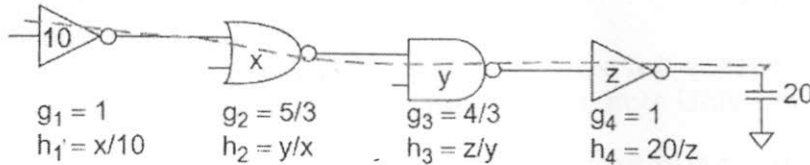


Figure Q5a

- b) Write gate level diagram for D flip flop with asynchronous set and reset inputs and explain the working.
6. a) Estimate rise time and fall time propagation delay of 3-input NAND gate if the output is loaded with h identical NAND gates?
- b) Explain the salient features of CVSL taking 2 input OR/NOR gate as an example, along with the general block diagram.

Unit – III

7. a) Write the (i) dynamic (ii) domino CMOS circuit for $Y = \overline{ABC + D + E}$ and explain the working.
- b) Explain the Built-In-Self-Test (BIST) technique with relevant diagram.
8. a) With neat diagram, explain the working of dynamic CMOS logic in precharge and evaluate modes.
- b) Explain stuck-at-faults, short circuit and open circuit faults with examples.

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
